

Wenyu Chiou

Artificial Intelligence Evaluation & Behavioral Simulation Research Engineer

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Research engineer building and evaluating artificial intelligence (AI) systems, including large language model (LLM) applications, agentic workflows, and behavioral simulations. Designed and fielded a **937-household** human-behavior survey and built a **52,141-household** coupled simulation; develops reproducible evaluations of whether LLM agents reproduce observed decision pathways across social groups.

Open to Summer 2027 internships in AI research, LLM evaluation, agent development, and computational social science.

Work authorization: F-1 student; CPT-eligible for internships.

— EXPERIENCE

Graduate Research Assistant — Complex Adaptive Water Systems Lab, Lehigh University

Aug 2024 – present

- Designed a human-versus-LLM evaluation study of **937** household responses (marginalized **n=148**; non-marginalized **n=789**) using multi-group structural equation modeling (SEM) and label-blind socioeconomic personas; implemented the redesigned elicitation with **13 regression tests** and piloted Gemma 3 4B and Llama 3; planned submission to *Progress in Disaster Science*.
- Designed and fielded a **937-household** flood-adaptation survey; used SEM to identify social-group pathway differences and Bayesian calibration to translate observed pathways into agent parameters.
- Built **FLOODABM**, a **52,141-household** coupled agent-based × catastrophe simulation across 27 census tracts and 2011–2023; simulated outcomes were validated against OpenFEMA insurance-claims data over 50 Monte Carlo runs.
- Designed and substantially completed experiments for the **Water Agent Governance Framework (WAGF)**, a six-stage governance and evaluation pipeline checking LLM-agent decisions against domain rules and behavioral theory before state updates; research software in preparation for release.

Independent Research Engineering & Open Source

2024 – present

- Built an **evaluation suite** for research-hub’s literature search that checks retrieval recall against expected papers; retained observed failures as explicit audit baselines instead of deleting them.
- Shipped **research-hub**, a local-first research tool with Model Context Protocol (MCP) and command-line interfaces (CLIs); published **research-hub-pipeline v1.1.1** on PyPI, tested across Windows/macOS/Linux, and contributed **18 merged pull requests** to third-party open source.
- Created and self-hosted [awesome-agentic-ai-zh](#), a trilingual agentic-AI learning roadmap with **240+ curated resources**, **23 hands-on exercises**, and **77 MCP/Skill entries**; reached **5K+ GitHub stars** and **700+ forks**.

— SELECTED RESEARCH OUTPUTS

Yang, Y. C. E., & Chiou, W. (2026). Leveraging Large Language Models for Agent-Based Simulation of Human–Water System Interactions. *Water Resources Research*, 62(6), e2025WR042111. DOI [Published](#)

Chiou, W., et al. Household Flood Adaptation and Financial Outcomes: A Coupled Human–Flood Modeling Analysis of Homeowners and Renters. *Water Resources Research*. [Under revision following peer review](#)

Chiou, W., et al. Flood Adaptation Decision Variables and Pathways across Social Groups: A Comparison between Human Survey Responses and Responses Generated by Large Language Models. Planned submission to *Progress in Disaster Science*.

— SELECTED CONFERENCE PRESENTATIONS

International Society for Data Science and Analytics (ISDSA) 2026 Annual Meeting — “What Makes AI Believable?” Comparing AI and human responses in mental-health risk assessment; recorded presentation.

American Geophysical Union (AGU) Fall Meeting 2025 — Poster NH41E-0449: “Modeling Long-Term Household Flood Adaptation under Social Heterogeneity: A Coupled Agent-Based Modeling Framework.”

The 2nd International Sociohydrology Conference (ISHC) 2025, Tokyo — Oral Presentation 38-03: “Investigating the Decision-Making Process and Causal Relationships between Flood Risk Perception and Adaptation Actions.”

— EDUCATION

Ph.D., Civil & Environmental Engineering, Lehigh University

2024 – present · expected 2028

M.S., Hydrological and Oceanic Sciences, National Central University

Sep 2021 – Jun 2023

B.S., Earth Sciences, National Central University

Sep 2017 – Jun 2021

— METHODS & TECHNICAL SKILLS

LLM / agent systems: LLM application & agent development · AI-system evaluation · evaluation infrastructure · prompt/context engineering · retrieval-augmented generation (RAG) · memory · tool use · multi-agent workflows · MCP

Methods: computational social science · behavioral simulation · experimental design · agent-based modeling (ABM) · structural equation modeling (SEM) · Bayesian calibration

Tools: Python · R · pytest / continuous integration (CI) · Git · OpenAI / Anthropic APIs · Ollama · Windows / POSIX